Sadia Binte Alam
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Education:

- **Master of Science (MS)**, Department of Biochemistry and Molecular Biology

University of Chittagong, Chattogram, Bangladesh

(July 2025-Present; Expected Completion December 2026)

Major Coursework: Genetics, Molecular Diagnostics, Forensic Technology, Clinical Biochemistry and Research, Drug Discovery and Development, Stem Cell Biology, Nanobiotechnology.

- **Bachelor of Science (B.Sc.),** Biochemistry and Molecular Biology, 2020-2025

CGPA: 3.83/4.00 (Class Rank: 2nd out of 42 students)
University of Chittagong, Chittagong, Bangladesh

Research Experience:

Laboratory of Molecular and Structural Immunology

Research Assistant (40hours/week); (Jan 2024-Present)

Principal Investigator and Advisor: Professor Dr. M. Monirul Islam

Department of Biochemistry and Molecular Biology, University of Chittagong

- **Thesis Ongoing:** Strategizing Mix and Match Vaccination Scheme on *E. coli* expressed SARS-CoV-2 RBDs; Wuhan, Omicron BA.5 and JN.1.
- As part of my thesis project, I am investigating the priming and booster dose response in Swiss albino mice groups. This study involves the utilization and optimization of techniques such as Indirect ELISA and qRT-PCR. In addition, I have experience in handling mice.
- Data processing and statistical evaluation are conducted using Microsoft Excel and GraphPad Prism.
- I also worked on a research project evaluating immune responses to wild-type Japanese encephalitis virus (JEV), assessing effects of specific viral mutations and adjuvant formulations.

Kuroda Laboratory

Special Research Student; (September 16, 2026 - December 16, 2026)

Supervisor: Yutaka Kuroda

Department of Biotechnology and Life Science, Graduate School of Engineering, Tokyo University of Science and Technology, Tokyo, Japan

- Expression and Purification of H3N2-C5I and Felline Type II RBD
- Evaluation of Immunogenicity of H3N2-C5I and H3N2 wild type in Balb/C mice model

Research Interest:

- Viral immunology and molecular mechanisms of host-pathogen interaction
- Structural vaccinology with emphasis on SARS-CoV-2 RBD-based subunit vaccines
- Development and evaluation of murine models for vaccine studies
- Protein Expression and Purification
- Neuroscience and Neurobiology

Additional Experience:

- Worked as a Project Management Secretary at a science club: Chittagong University Scientific Society
- Awarded champion for Best Poster Presentation, Idea for Innovation 2.0, organized by Biochemistry and Molecular Biology Career Club
- Participated in the SCIENSPUR Cell Biology Cohort 2022, a program initiated by postdoctoral researchers and biotech industry experts to deliver STEM education via online platforms

Conferences:

- A.R. Mazumder**,** M.M. Fuad, **S.B. Alam**, R.A. Erin, M.M. Islam; *Multivalent SARS-CoV-2 RBD Immunization Generates Cross-Reactive Antibodies in Mice Model,* International Biotechnology Conference (IBC), Brac University, Dhaka, Bangladesh, June 21, 2025(p.189-190).

- Attended International Conference and Bioscience Carnival at Rangamati Science and Technology University, Rangamati, Chittagong, Bangladesh

Awards and Fellowships:

National Science and Technology Fellowship 2025-2026, by The Ministry of Science and Technology, Bangladesh.

HEAT-11009-CU MS Research Scholarship under Higher Education Acceleration and Transformation (HEAT) Project.

JASSO Scholarship under JASSO Student Exchange Support Program, affiliated with MEXT, Japan.

Skills:

Research based: ELISA, qRT-PCR, DNA/RNA extraction, Nanodrop Spectrophotometer, Mouse Handling, Bacterial Transformation, Protein Expression and Purification, SDS-PAGE, RP-HPLC

Software: Microsoft Office, Basic R Programming

Languages: Bengali (native), English (professional proficiency)

Others: Sample collection (field-level), Literature Review, Scientific writing, Lab Management, Event Management